Lab: Perception as Prediction

> **Learning Goal:** Experience and analyze the constructive, predictive nature of perception.

Part 1: Experiencing Illusions

Instructions: Look up and try each of the following illusions (search online for images/videos), then explain each using the predictive processing framework.

1. **The Hollow Mask Illusion**: Search for a rotating hollow mask video. What does your brain predict? What is the prediction error?
2. **The Checker Shadow Illusion** (Adelson): Squares A and B are the same shade of gray. Why does your brain perceive them as different?
3. **The McGurk Effect**: Search for a video demonstration. What happens when audio "ba" is paired with visual "ga"?

For each, state: (a) the prediction, (b) the sensory data, (c) the prediction error, (d) how the brain resolves the conflict.

Write your response here

Part 2: Precision in Action

> **Learning Goal:** Explore how confidence weighting shapes perception.

Scenario analysis: For each situation, explain how precision changes and what effect it has on perception.

1. You hear a strange noise at night when home alone vs. when with friends.
2. You're reading a text message while walking in bright sunlight vs. in a dimly lit room.

3. A doctor examines an X-ray of a familiar pathology vs. a rare and ambiguous case.

Write your response here

Part 3: Prior Expectations Shape Perception

> **Learning Goal:** Demonstrate that identical sensory data produces different percepts depending on priors.

Exercise: Consider the ambiguous text: "I saw her duck."

1. List at least two different interpretations.
2. What prior knowledge changes which interpretation you select?
3. Find another example of an ambiguous sentence or image and analyze it the same way.

Write your response here

Part 4: When Prediction Goes Wrong

> **Learning Goal:** Analyze cases where the predictive system produces distorted perception.

Case studies: For each condition, explain how predictive processing might account for the symptoms.

1. **Charles Bonnet Syndrome**: Blind or visually impaired people sometimes "see" vivid hallucinations of faces, animals, or patterns.
2. **Phantom Limb Pain**: Amputees often feel pain or sensation in a limb that no longer exists.

3. **Déjà Vu**: The strong feeling that you have experienced a current situation before.

Write your response here

Part 5: Reflection

In 150 words, consider: If all perception is prediction, what is the relationship between perception and imagination? Are they the same process running in different "modes"?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Illusion analysis	Prediction vs. sensory data
2	Precision reasoning	Confidence weighting
3	Linguistic analysis	Prior expectations and ambiguity
4	Clinical reasoning	Prediction errors in pathology
5	Philosophical reflection	Perception vs. imagination